

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: COMPUTER VISION WITH OPENCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

Assessment Tool Weightage: 10%

Technical Presentation Topics

1. Deliver a technically accurate presentation explaining the introduction, key features, importance, and levels of Computer Vision, including a clear problem context where these concepts are applied in real-world systems. Use appropriate visual tools (diagrams/flowcharts), ensure structured communication, and demonstrate professional delivery.
2. Present a technically accurate analysis of pixel representation, clearly identifying challenges in pixel-based image processing, using appropriate visual tools, and delivering insights in a structured and professional manner.
3. Analyze spatial resolution in images, defining real-world problems where resolution impacts performance, supporting with visual comparisons, and communicating findings clearly and professionally.
4. Explain intensity resolution through problem-based scenarios, using suitable graphical tools to demonstrate its effects and presenting the analysis with clarity and technical depth.
5. Present grayscale image representation, identifying practical limitations, analyzing them using examples or tools, and ensuring clear and professional communication.
6. Compare CCD and CMOS sensors with technical accuracy, defining engineering trade-offs, using visual comparison tools, and presenting findings in a structured and professional format.

7. Analyze challenges in image sensing systems, clearly defining acquisition-related problems, applying appropriate analytical methods, and supporting explanations with diagrams or models.
8. Investigate the impact of environmental conditions on image acquisition, identifying real-world problems, using visual evidence/tools, and presenting insights clearly.
9. Analyze under-sampling in digital images, defining the problem, demonstrating effects using graphs or simulations, and communicating results effectively.
10. Investigate over-quantization effects, identifying limitations, using visual tools for demonstration, and presenting findings with clarity and professionalism.
11. Explain the relationship between sampling and resolution, defining the underlying problem, using analytical/visual tools, and ensuring structured communication.
12. Present a comparative technical analysis of Image Processing and Computer Vision by clearly defining the problem scope of each domain, using suitable visual aids (comparison tables/diagrams), and communicating insights in a clear, organized, and professional manner.
13. Explain digital image fundamentals with technical depth, including pixel representation, spatial and intensity resolution, and grayscale images. Define practical problems related to image quality, support explanations with visual tools (sample images/plots), and ensure clarity and professionalism in delivery.
14. Analyze the image acquisition process by identifying real-world challenges in capturing images, explaining components and sensor roles (CCD vs CMOS), and using diagrams or system models. Present findings clearly with effective visuals and professional communication.
15. Present a problem-driven explanation of sampling and quantization, including effects like under-sampling and over-quantization. Use graphs, simulations, or visual examples as tools to support analysis, and communicate results in a structured and engaging manner.
16. Explain image formation using appropriate technical models (pinhole camera, perspective projection), identifying problems such as distortion and illumination effects. Use diagrams or simulations, and present the content clearly with professional delivery.
17. Analyze real-world problems across domains (healthcare, surveillance, autonomous vehicles, agriculture, retail) and explain how Computer Vision provides solutions. Use case studies, visuals, or data-driven tools, and present findings in a clear and engaging format.
18. Develop an integrated presentation linking acquisition, resolution, sampling, and quantization to real-world system performance. Clearly define the problem context, use conceptual frameworks/diagrams, and communicate insights effectively with professionalism.
19. Present a complete pipeline analysis from real-world scene to digital image and processing workflow. Identify key challenges, use flow diagrams or system

models, and deliver the content with clarity, technical accuracy, and professional engagement.

20. Investigate and present key factors affecting image quality (acquisition conditions, resolution, noise, distortion), clearly defining the problem and analyzing it using appropriate tools (visual comparisons, charts). Ensure structured communication and professional presentation.
21. Analyze how image acquisition quality influences Computer Vision outcomes, defining system-level problems, using workflow diagrams, and presenting insights professionally.
22. Present a detailed Computer Vision workflow, identifying bottlenecks, analyzing them logically, and supporting with clear visual tools.
23. Investigate the impact of image formation models on application accuracy, defining related problems and using diagrams or simulations for explanation.
24. Analyze image distortion and its effect on processing, identifying causes, using visual examples/tools, and presenting findings clearly.
25. Explain the effect of illumination on image formation, defining real-world challenges, using graphical tools, and communicating insights effectively.
26. Present perspective projection in vision systems, identifying application-based problems and explaining them using diagrams and structured communication.
27. Analyze the relationship between image quality and application performance, defining the problem, using comparative tools, and presenting findings professionally.
28. Investigate challenges in real-world scene capture, defining constraints, analyzing solutions, and supporting with system diagrams.
29. Explain interactions between stages of the Computer Vision pipeline, identifying dependencies and presenting them using integrated visual frameworks.
30. Analyze error propagation in a Computer Vision pipeline, defining the problem, using case-based or visual tools, and presenting insights clearly.
31. Present a technical analysis of Computer Vision in healthcare, identifying real-world problems, using case studies/tools, and delivering insights professionally.
32. Analyze surveillance applications using Computer Vision, defining challenges, using system models or visuals, and presenting findings clearly.
33. Investigate Computer Vision in autonomous vehicles, identifying technical problems, applying analytical methods, and using appropriate visuals.
34. Explain Computer Vision applications in agriculture, identifying practical problems and presenting solutions using structured tools and visuals.
35. Analyze retail applications of Computer Vision, defining system challenges, using workflow diagrams, and communicating insights professionally.
36. Evaluate trade-offs between image resolution and processing cost, defining the problem, using analytical comparisons/tools, and presenting findings clearly.
37. Analyze the impact of noise on image quality and system accuracy, identifying the problem, using graphical tools, and presenting structured insights.
38. Investigate factors affecting image clarity, defining real-world issues, analyzing them using appropriate tools, and communicating results effectively.

39. Evaluate different image acquisition setups, defining performance problems, using comparative visuals/data, and presenting findings professionally.
40. Present a critical analysis of limitations in Computer Vision systems, identifying key technical challenges, proposing solutions, and communicating insights clearly and professionally.
41. Present the evolution of Computer Vision, identifying key technological milestones, using timelines/visuals, and explaining their impact on modern systems.
42. Analyze the role of feature extraction in Computer Vision, defining practical problems, using diagrams, and presenting structured insights.
43. Explain hierarchical levels of vision processing, identifying real-world applications at each level, supported by flow diagrams.
44. Investigate challenges in transitioning from image processing to Computer Vision, defining system-level problems and presenting structured comparisons.
45. Present a technical explanation of human vision vs computer vision, identifying differences using comparative diagrams and real-world examples.
46. Analyze pixel density variation across devices, defining performance impacts and presenting visual comparisons.
47. Explain aliasing effects in digital images, demonstrating with graphical tools and structured explanation.
48. Investigate the relationship between pixel size and image quality, using real-world examples and visual aids.
49. Analyze trade-offs between spatial resolution and storage requirements, using comparative charts and structured communication.
50. Present challenges in multi-resolution image representation, using diagrams and application-based examples.
51. Explain noise sources in image sensors, identifying real-world challenges and presenting mitigation strategies with diagrams.
52. Analyze dynamic range in image sensing systems, defining practical problems and using graphical tools for explanation.
53. Investigate lens effects in image acquisition, identifying distortion problems and presenting solutions using diagrams.
54. Present a comparative study of analog vs digital image acquisition, using structured visuals and examples.
55. Analyze real-time acquisition constraints in embedded vision systems, using system models and structured explanation.
56. Analyze aliasing effects in video streams, defining problems and demonstrating with simulations or visual examples.
57. Explain quantization noise in digital imaging, identifying its impact and presenting graphical analysis.
58. Investigate optimal sampling strategies for high-detail applications, using case studies and diagrams.
59. Present trade-offs between sampling rate and computational cost, using comparative visual tools.

60. Analyze adaptive quantization techniques, defining problems and presenting solutions with structured visuals.
61. Explain camera calibration in image formation, identifying distortion issues and presenting solutions using diagrams.
62. Analyze perspective distortion in real-world images, using visual examples and structured explanation.
63. Investigate illumination models in Computer Vision, defining challenges and presenting graphical interpretations.
64. Present depth perception using image formation principles, using diagrams and application examples.
65. Analyze geometric transformations in image formation, using visual tools and structured communication.
66. Present an end-to-end vision system pipeline for surveillance, identifying stages and challenges using flow diagrams.
67. Analyze bottlenecks in real-time Computer Vision systems, using system-level diagrams and structured explanation.
68. Investigate scalability challenges in large-scale vision systems, defining problems and presenting solutions.
69. Present data flow in a Computer Vision pipeline, identifying dependencies and using structured visuals.
70. Analyze system-level trade-offs between accuracy and efficiency, using comparative tools and structured explanation.
71. Present Computer Vision applications in smart cities, identifying problems and solutions using case studies.
72. Analyze healthcare imaging systems, defining diagnostic challenges and presenting structured solutions.
73. Investigate Computer Vision in robotics, identifying perception challenges and presenting system models.
74. Present industrial automation using vision systems, defining problems and using workflow diagrams.
75. Analyze security applications using Computer Vision, identifying challenges and presenting structured insights.
76. Analyze factors affecting image sharpness, using visual comparisons and structured explanation.
77. Investigate contrast enhancement challenges, defining problems and presenting solutions with graphs.
78. Present noise reduction techniques in imaging systems, using visual examples and structured communication.
79. Analyze blur effects in images, identifying causes and presenting solutions using diagrams.
80. Investigate compression effects on image quality, defining trade-offs and presenting analysis.
81. Analyze the impact of sensor limitations on Computer Vision accuracy, using system models and structured explanation.

82. Investigate real-time processing constraints in vision systems, defining bottlenecks and presenting solutions.
83. Present error sources in image acquisition pipelines, using diagrams and structured analysis.
84. Analyze system robustness under varying environmental conditions, using case studies and visual tools.
85. Investigate reliability challenges in Computer Vision systems, defining problems and presenting solutions.
86. Compare different imaging modalities (visible vs infrared), defining applications and using structured visuals.
87. Analyze trade-offs between hardware and software optimization in vision systems, using comparative tools.
88. Present evaluation metrics for image quality, using graphs and structured explanation.
89. Compare different acquisition pipelines in real-world systems, using diagrams and structured communication.
90. Analyze system performance under varying resolution settings, using comparative analysis.
91. Design an image acquisition system for low-light conditions, defining constraints and presenting solutions.
92. Develop a strategy to improve image quality in surveillance systems, using structured frameworks.
93. Analyze design challenges in mobile vision systems, using system diagrams and structured explanation.
94. Present a solution for reducing noise in imaging pipelines, using visual tools and structured reasoning.
95. Investigate optimization strategies for real-time vision systems, defining constraints and presenting solutions.
96. Present a case study on image acquisition failures in real systems, analyzing causes and solutions.
97. Analyze real-world challenges in drone imaging systems, using diagrams and structured explanation.
98. Investigate imaging challenges in underwater environments, defining problems and presenting solutions.
99. Present a case study on medical imaging errors, analyzing causes and improvements.
100. Analyze failures in autonomous vision systems, identifying causes and presenting structured solutions.
101. Present integration of multiple sensors in vision systems, defining challenges and using diagrams.
102. Analyze workflow optimization in image processing pipelines, using structured tools.
103. Investigate synchronization issues in multi-camera systems, defining problems and presenting solutions.

104. Present data handling challenges in large-scale vision systems, using structured diagrams.
105. Analyze system integration challenges in real-time applications, using case-based explanation.
106. Present emerging trends in Computer Vision, using structured analysis and real-world examples.
107. Analyze the impact of AI on modern vision systems, using case studies and diagrams.
108. Investigate edge computing in Computer Vision, defining challenges and presenting solutions.
109. Present advancements in imaging sensors, using comparative analysis and visuals.
110. Analyze future challenges in large-scale vision deployment, using structured reasoning.
111. Present a holistic evaluation of a Computer Vision system, identifying strengths, weaknesses, and improvements.
112. Analyze limitations of current imaging technologies, using structured comparison and visual tools.
113. Investigate ethical challenges in Computer Vision applications, using case studies and structured explanation.
114. Present a critical analysis of data dependency in vision systems, using structured reasoning.
115. Analyze scalability issues in real-world deployments, using diagrams and structured communication.
116. Design a complete vision-based system for a real-world application, presenting pipeline, challenges, and solutions.
117. Present a detailed system architecture for an autonomous vision system, using diagrams and structured explanation.
118. Analyze end-to-end performance of a vision system, identifying bottlenecks and presenting improvements.
119. Develop a conceptual framework linking acquisition, processing, and application performance, using diagrams.
120. Present a comprehensive analysis of Computer Vision system design, integrating all stages with real-world examples.

Code of Conduct:

I Bhavya PS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

CO1 ASSESSMENT TOOL 3 –TECHNICAL PRESENTATION

TOPIC_1: Impact of Image Formation Models on Application Accuracy

DSA0204 - COMPUTER VISION WITH OPENCV

Faculty :
Dr.senthil

Presented by
bhavya ps

INTRODUCTION

- *Impact of Image Formation Models on Application Accuracy*
- Image formation defines how scenes are captured
- Accuracy depends on correct modeling

Types of Models



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graph TD; A[Types of Models] --> B[Geometric: pinhole camera, perspective projection]; A --> C[Radiometric: light intensity, reflectance models]
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Geometric: pinhole camera,
perspective projection

Radiometric: light intensity,
reflectance models

Problems in Image Formation

- Noise, blur, occlusion
- Sensor limitations
- Environmental interference

Impact on Applications

- Misdiagnosis in medical imaging
- Reduced recognition in AI systems
- Errors in remote sensing

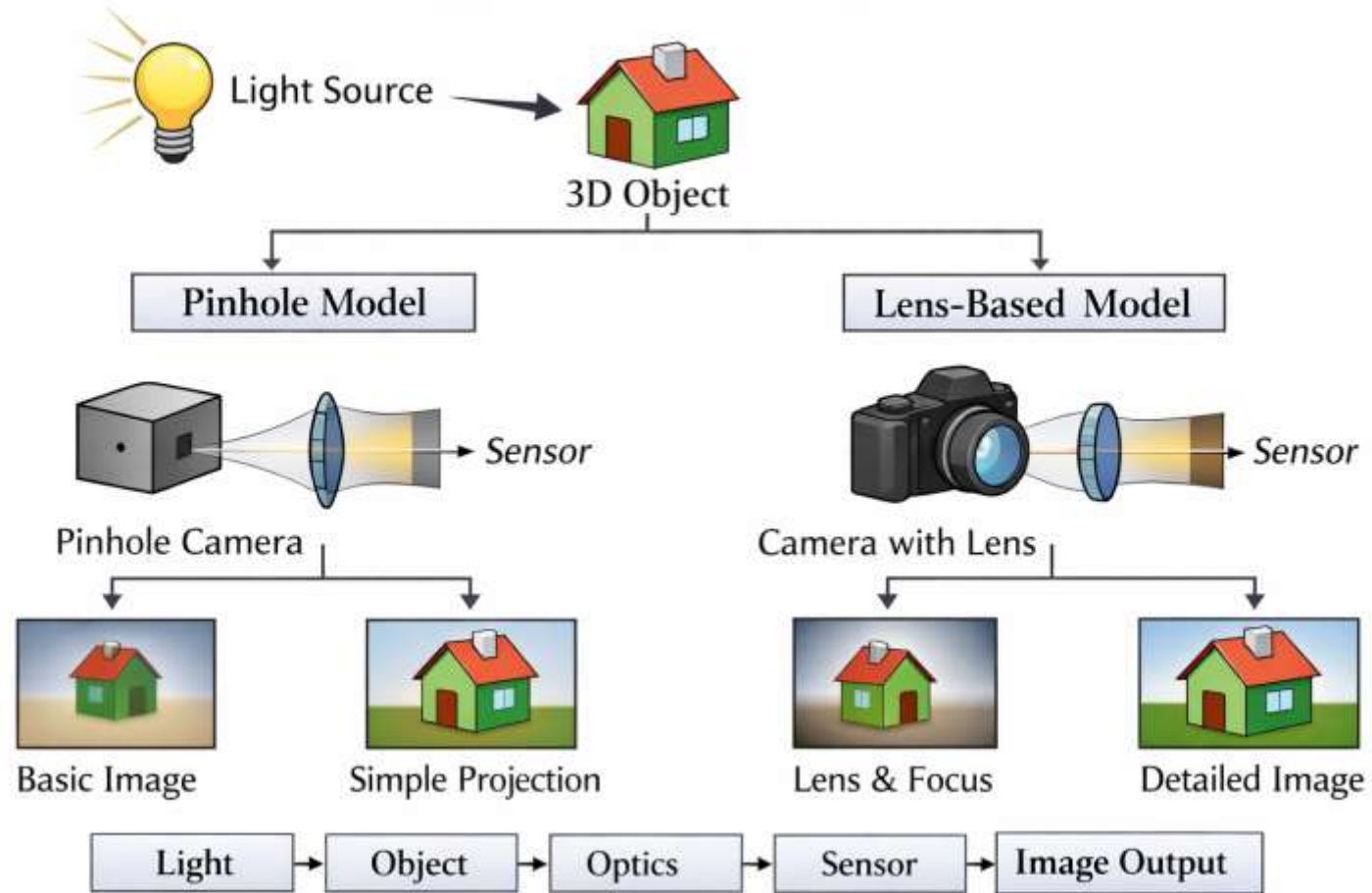
Diagram – Image Formation Pipeline

- Diagram – Image Formation Pipeline

Light Source → Object → Lens → Sensor → Image Output

- Shows how geometric and radiometric models interact
- Comparison:
 - Pinhole camera model → simple projection, less distortion
 - Lens-based model → realistic capture, but introduces blur/distortion

Image Formation Pipeline



References

- R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Cham, Switzerland: Springer, 2022.
- R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.
- D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed. Pearson, 2012.

Topic_2: Image Distortion and Its Effect on Processing

Introduction

- *Image Distortion and Its Effect on Processing*
- Distortion = deviation from true geometry
- Critical in computer vision tasks

Types of Distortion

- Geometric: barrel, pincushion
- Radiometric: uneven illumination
- Motion blur

Causes of Distortion

- Lens imperfections
- Heat/turbulence in environment
- Motion of camera/object

Effect on Processing

- Reduced recognition accuracy
- Misclassification in ML models
- Errors in measurements

Findings

- Calibration is essential
- Modern correction: deep learning, adaptive filters

Distorted Image



Corrected Image



References

- Z. Zhang, "A Flexible New Technique for Camera Calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, Nov. 2000.
- R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Cham, Switzerland: Springer, 2022.

Topic_3: Effect of Illumination on Image Formation

Introduction

- Effect of Illumination on Image Formation
- Illumination defines visibility and clarity
- Real-world relevance in imaging

Illumination Models

- Ambient, diffuse, specular reflection
- Lambertian vs Phong models

Challenges in Real World

- Shadows, highlights, glare
- Low-light conditions
- Uneven illumination

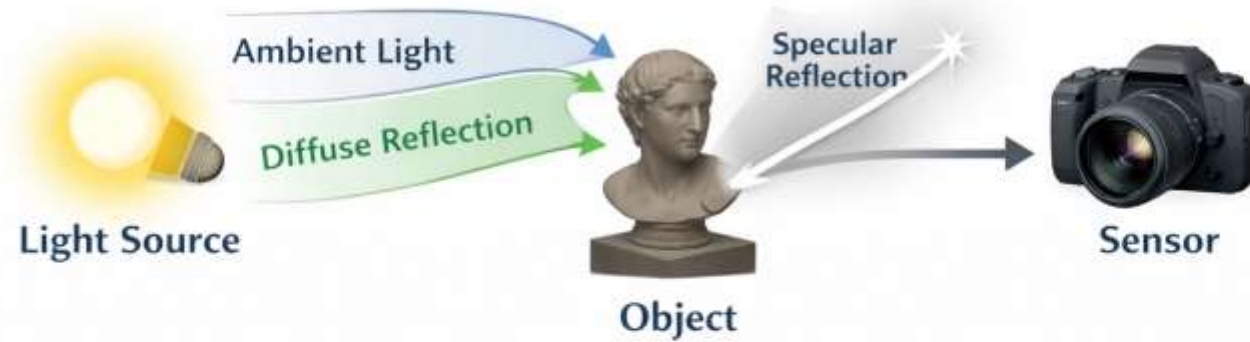
Effect on Applications

- Face recognition under varying light
- Autonomous driving at night
- Medical imaging under uneven lighting

Insights

- Illumination-invariant models needed
- AI-based enhancement: HDR, adaptive exposure

Illumination Effect on Imaging



Low Light



Bright Light



Balanced Light



References

- B. K. P. Horn, *Robot Vision*. MIT Press, 1986.
- R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.
- D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed. Pearson, 2012.

THANK YOU